

# ELI — The Complete User Manual

Plain-English Edition · with flowcharts

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This manual assumes **zero** technical background. You talk to ELI in normal English — by **typing** in the chat box or **speaking** — and it does things on your computer, answers questions, remembers what matters to you, and learns your routines. Everything runs **on your own machine**; nothing is sent to the cloud.

**The one rule worth knowing:** ELI is **offline by default**. It only touches the internet when *you* ask for something that needs it (a web search, the news, downloading a model). Your conversations, files, and memories never leave your computer.

How to read this manual: - **Just want to use it?** Read §1–§5 and the cheat sheet (§18). That's enough to be productive. - **Want the whole picture?** Read straight through — every feature is here, in plain terms. - **[Anyone]** / **[A bit technical]** / **[Advanced]** **tags** at the start of some sections mean *anyone* / *a little technical* / *advanced*.

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## 1. What ELI is (in one minute)

ELI is a **personal AI assistant that lives on your computer**. Think of it as a capable helper that can:

- **Talk with you** like a knowledgeable person — by text or voice.
- **Operate your computer** — open apps, play music, manage windows, type for you.
- **Look at things** — your screen, images, PDFs, spreadsheets — and explain them.
- **Write for you** — documents, notes, code, scripts.
- **Remember** what matters about you, and **learn** your daily routines.
- **Act on its own** when helpful — a morning briefing, a heads-up, a suggestion.

The big difference from cloud assistants: **it's yours**. It runs locally, stays private, has no subscription, and you can even retrain its “voice” yourself (\$14).

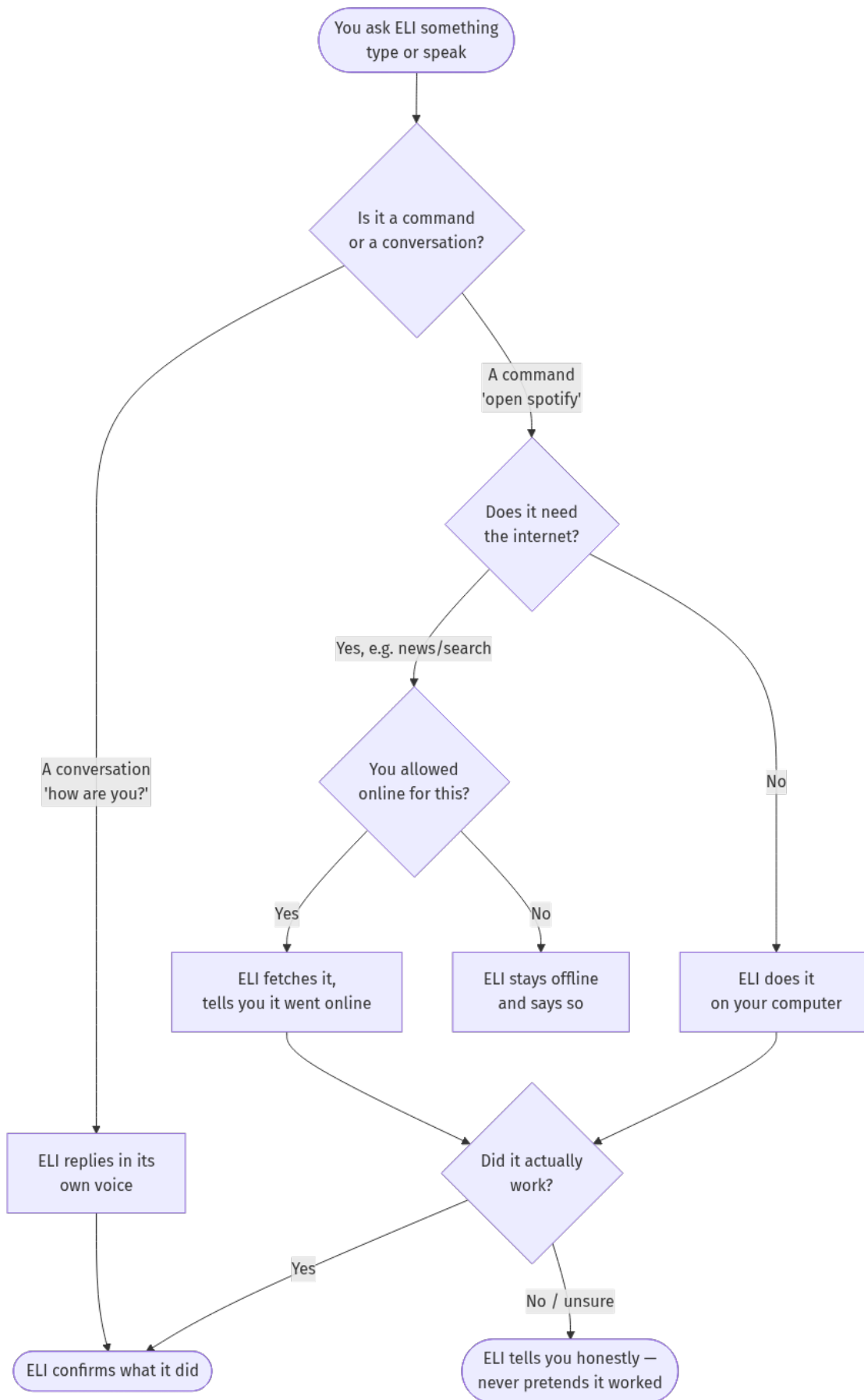
Here's the whole thing at a glance:



## 2. How ELI works when you ask it something

[Anyone] Anyone — *this is the mental model that makes everything else click.*

When you say or type something, ELI follows the same honest loop every time:



**The promise in that diagram:** ELI **never fakes an action**. If it says it played a song or fixed a file, it really did — and if it couldn't, it says so plainly instead of bluffing. This is built into the software, not just good manners.

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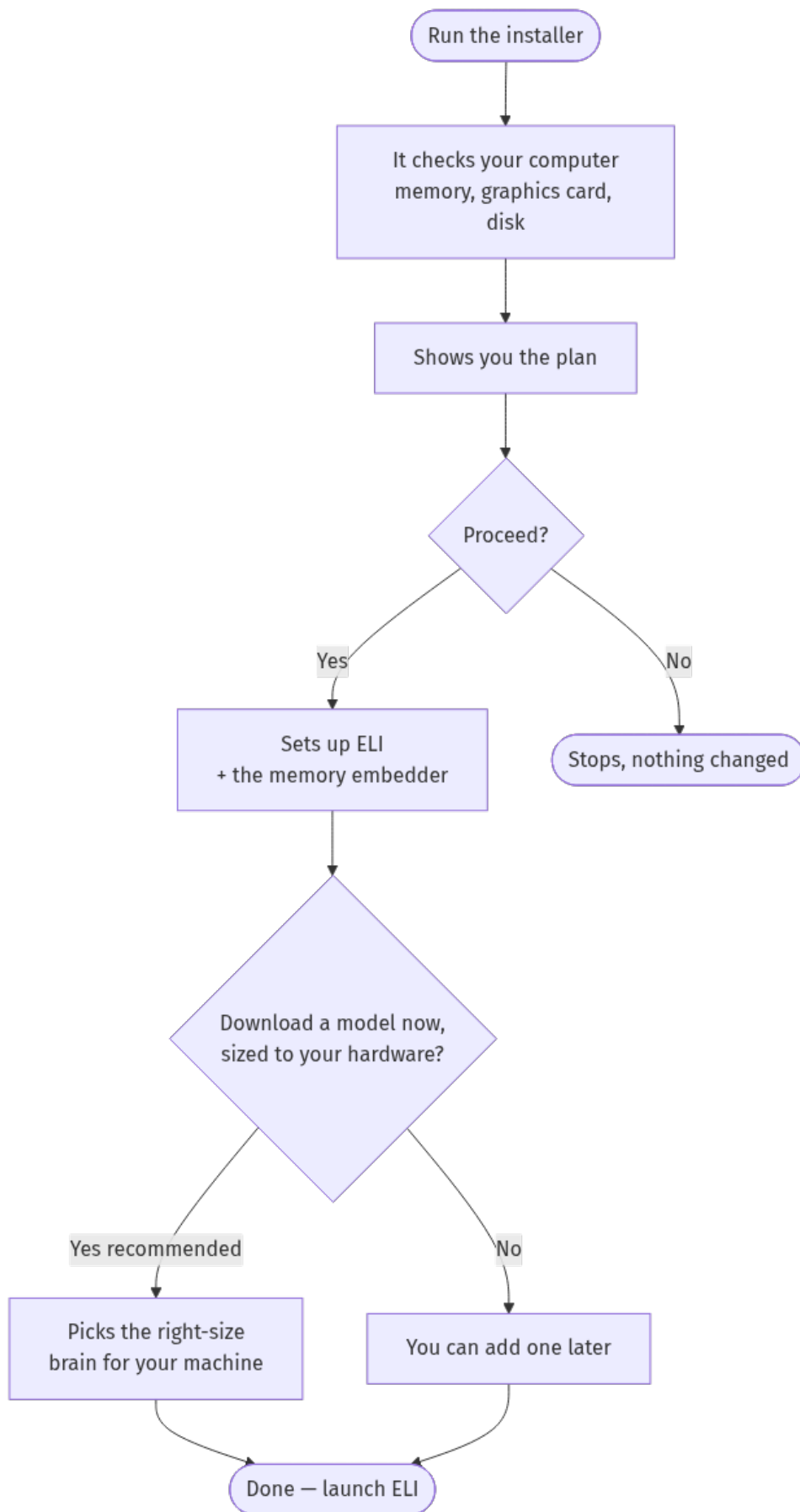
### 3. Getting started

[A bit technical] *A little technical, but only once.*

#### Installing

You run a single installer and answer a couple of friendly questions.

- **Windows:** run `install.ps1`
- **Mac & Linux:** run `install.sh`



The installer also fetches a tiny **embedder** model automatically (it powers ELI's memory), so you don't have to think about it. Everything ELI needs ends up in place — no extra steps.

## First launch — the quick interview

The very first time you open ELI, it doesn't know you yet, so it asks **three short questions** (your name, what you do, how you like to be spoken to). Ten seconds, and you can **skip it** entirely. From there, ELI builds its understanding of you naturally as you talk.

## Opening ELI

- **Desktop app (full experience):** launch **ELI Pro** from your apps menu, or run `scripts/eli_launch.sh` (Mac/Linux) / the desktop shortcut (Windows).
  - **From your phone:** see §13.
- 

## 4. Your first hour with ELI

[Anyone] A gentle on-ramp. Try these in order — each builds confidence.

1. **Say hello.** Type how are you? — get a feel for ELI's voice.
2. **Ask what it can do.** what can you do? lists its capabilities.
3. **Open something.** open the file manager or open firefox.
4. **Play music.** play lo-fi on YouTube, then pause, then skip.
5. **Look at your screen.** what's on my screen?
6. **Tell it a fact.** remember that my dog's name is Rufus. Later: what's my dog's name?
7. **Set a reminder.** set a timer for 5 minutes.
8. **Get a briefing.** what's the news? (this one goes online — ELI will say so).
9. **Check on ELI itself.** what are you running on? shows its model and hardware.
10. **Go hands-free.** Say the wake word **“computer”**, then a request.

By the end you've touched conversation, control, vision, memory, reminders, the web, voice, and introspection — the whole surface.

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## 5. Talking to ELI

There are two ways, freely mixed:

- **Type** in the chat box and press Enter.
- **Speak** — click the microphone, or say the **wake word** (default **“computer”**) then your request. Change the wake word to anything (§11).

You don't need special phrasing. Talk normally:

“open spotify and play some lo-fi” “what's on my screen?” “remember that my sister's name is Anna” “set an alarm for 7am”

**Chaining:** ask for several things at once — > “close steam and set an alarm for 7am” > “open spotify then play Juicy by Notorious B.I.G.”

**If ELI isn't sure** what you meant, it asks — it won't guess wildly or pretend. And if a request needs the internet, it tells you before going online.

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## 6. A tour of the window

[Anyone] ELI Pro is organised into **tabs**. You'll mostly live in **Chat**; here's the rest in plain terms:

| Tab                   | What it's for                                                           |
|-----------------------|-------------------------------------------------------------------------|
| <b>Chat</b>           | Talk to ELI, by voice or text. Home base.                               |
| <b>Memory</b>         | Browse what ELI remembers about you; add or remove facts.               |
| <b>Habits</b>         | Routines ELI has learned; turn on/off, edit times, add your own.        |
| <b>Tasks</b>          | Scheduled & overnight jobs and their status.                            |
| <b>Proactive</b>      | Controls for ELI's "act on its own" behaviour.                          |
| <b>Report Builder</b> | Generate longer documents/reports with sources.                         |
| <b>Labs</b>           | Power-user tools — image generation, code tools, projects, evaluations. |
| <b>Settings</b>       | Model, hardware, voice, and privacy options (\$16).                     |

You rarely need the tabs directly — almost everything is reachable by **asking** in Chat: "show my habits", "what do you remember about me", "show background jobs", and ELI opens or reports the right thing.

## 7. What ELI can do — by everyday task

[Anyone] Real things you can say. ELI understands many phrasings — these are examples.

### Music & media

- "play lo-fi on YouTube" · "play Juicy by Notorious B.I.G. on Spotify"
- "pause" · "skip" · "next song" · "previous track" · "shuffle" · "repeat this song"
- "what's playing?" · "stop the music" · "skip the ad" (YouTube, when skippable)

### Apps & windows

- "open spotify" · "launch firefox" · "open the file manager" · "open github.com"
- "close chrome" · "focus the browser" · "switch to workspace 2"
- "tile windows" · "maximise the window" · "show desktop" · "next window"

### Volume, typing, mouse

- "volume up" · "mute" · "set volume to 30"
- "type hello world" · "press enter" · "left click" · "move the mouse up"

### Files & notes

- "create a file notes.txt" · "make a folder called projects" · "read notes.txt"
- "list the files in ~/Documents" · "write a note saying buy milk" · "search notes for budget"
- "what's in my clipboard?" · "summarise report.docx" · "convert report.md to pdf"

## Writing, documents & code

- “write a report on renewable energy” · “generate a document about X”
- “write a bash script to monitor the GPU” · “solve this: implement a function that ...”
- “fix the bugs in foo.py” · “examine eli/memory/memory.py for errors” → ELI scans, **offers** fixes, and applies them only if you say “yes, fix it”. · “show the diff”

## Screen, images, PDFs, spreadsheets

- “what’s on my screen?” · “find the submit button on screen” · “take a screenshot”
- “describe cat.jpg” · “read the text in image.jpg” (OCR)
- “summarise report.pdf” · “analyse data.csv” · “watch my screen” / “stop watching”

## Information

- “what time is it?” · “what’s the date?” · “weather in Wexford”
- “what’s the news?” / “catch me up” → a **synthesised** briefing, not a raw dump
- “search the web for X” (goes online — ELI tells you) · “gpu status” · “system stats”

## Reminders, timers, calendar, scheduling

- “set an alarm for 7am” · “set a timer for 10 minutes”
- “add an event tomorrow at 3pm” · “list my events” · “start a pomodoro”
- “research the best solar inverters overnight” · “build me a script at 2am” → **background jobs**
- “show background jobs” · “check job 5”

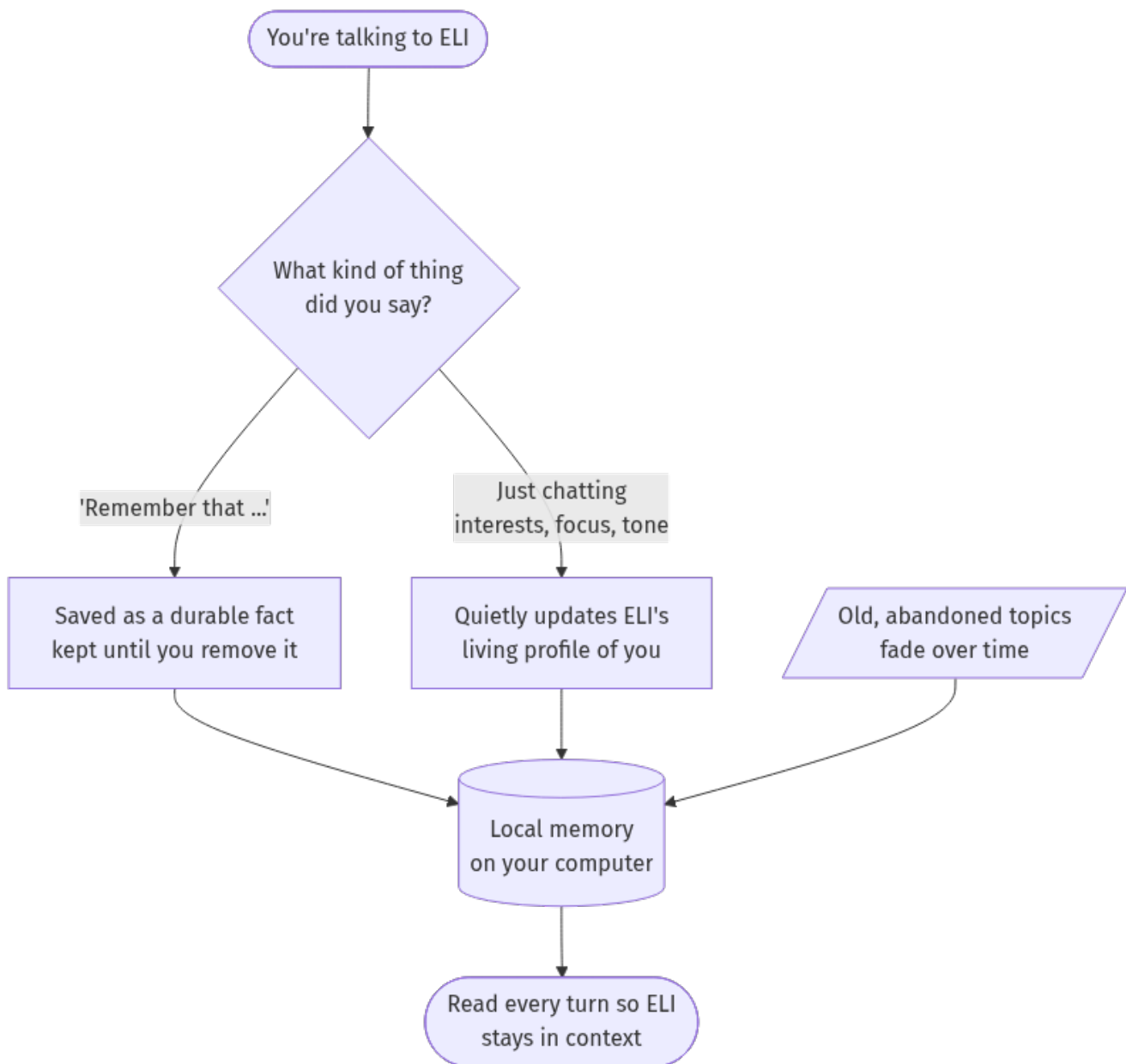
## Asking ELI about itself (honest introspection)

- “what are you running on?” (model, context, GPU) · “how does your memory work?”
  - “what do you know about me?” · “what have you been working on?” · “status”
- 

## 8. ELI remembers you (memory)

[Anyone] ELI builds a private, evolving understanding of **who you are** and reads it every time you talk, so you never repeat yourself.



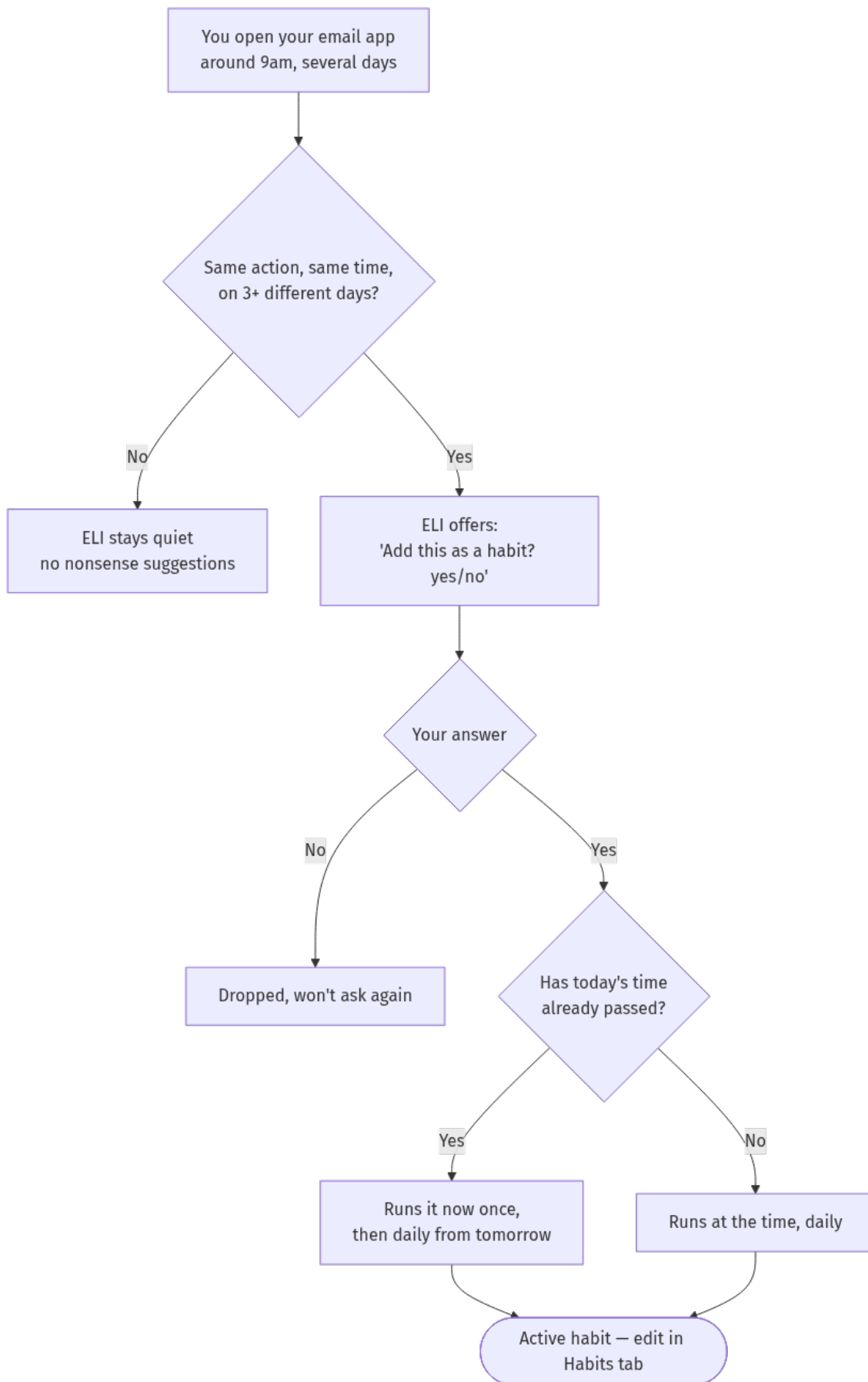


**Tell it something to keep:** “remember that my sister’s name is Anna” · “my name is Alex” **Ask what it knows:** “what do you remember about my car?” · “what do you know about me?” · “explain everything you know about me and where it’s stored” (a sourced, detailed answer).

Everything is stored **locally** in small private databases. Review and prune it in the **Memory** tab. A brand-new install is a **blank slate** — ELI knows nothing about you until you talk.

## 9. ELI learns your routines (habits)

[Anyone] If you do the same thing at roughly the same time on **several different days**, ELI notices and **offers** to automate it. It never activates anything without your “yes”.



Manage everything in the **Habits** tab (add, edit times, turn off), or say “show my habits”. The key safeguards: it only proposes routines you’ve **genuinely repeated across days**, and it **always asks first**.

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## 10. ELI helps before you ask (proactive)

[Anyone] With **proactive mode** on, ELI can do helpful things on its own — a **morning briefing**, a gentle suggestion, or a goal it thinks would help. It’s conservative and easy to control:

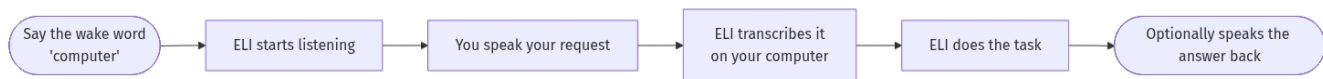
- “start proactive mode” / “stop proactive mode” · “proactive status”
- “morning report” / “give me my briefing” (news, weather, your agenda)
- “do you have any proposals for me?” / “what’s on your agenda?”

Schedule real unattended work too: > “research X overnight” · “generate tests for your code tonight”  
These appear in the **Tasks** tab and survive restarts.

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## 11. Voice, wake word, and dictation

[Anyone] ELI is fully usable hands-free.



- **Wake word:** say “**computer**” then your request. Change it: “change the wake word to athena”. ELI trains its listener on the new word automatically — works even over background music.
  - **Personalise to your voice:** “enroll my wake word” (records you saying it and adapts).
  - **Dictation:** “start dictation” / “stop dictating” — speak and ELI types into the active field.
  - **Transcribe a recording:** “transcribe audio.wav”
  - **ELI speaking back:** “say good morning”
  - **Teach ELI your voice & tone:** “train my voice” — it learns your pitch/energy and even emotion cues, then adapts how it speaks to you.
  - **If voice misbehaves:** “run voice diagnostics”.
- 

## 12. Seeing your screen and images (vision)

[Anyone] ELI can actually **look**:

- **Your screen:** “what’s on my screen?” · “what does this error say?” · “find the submit button”
- **Image files:** “describe cat.jpg” · “read the text in this image” (OCR)
- **Documents:** “summarise report.pdf” · “analyse data.csv” · “analyse the pdfs in that folder”
- **Continuous watching:** “watch my screen” (and “stop watching”)

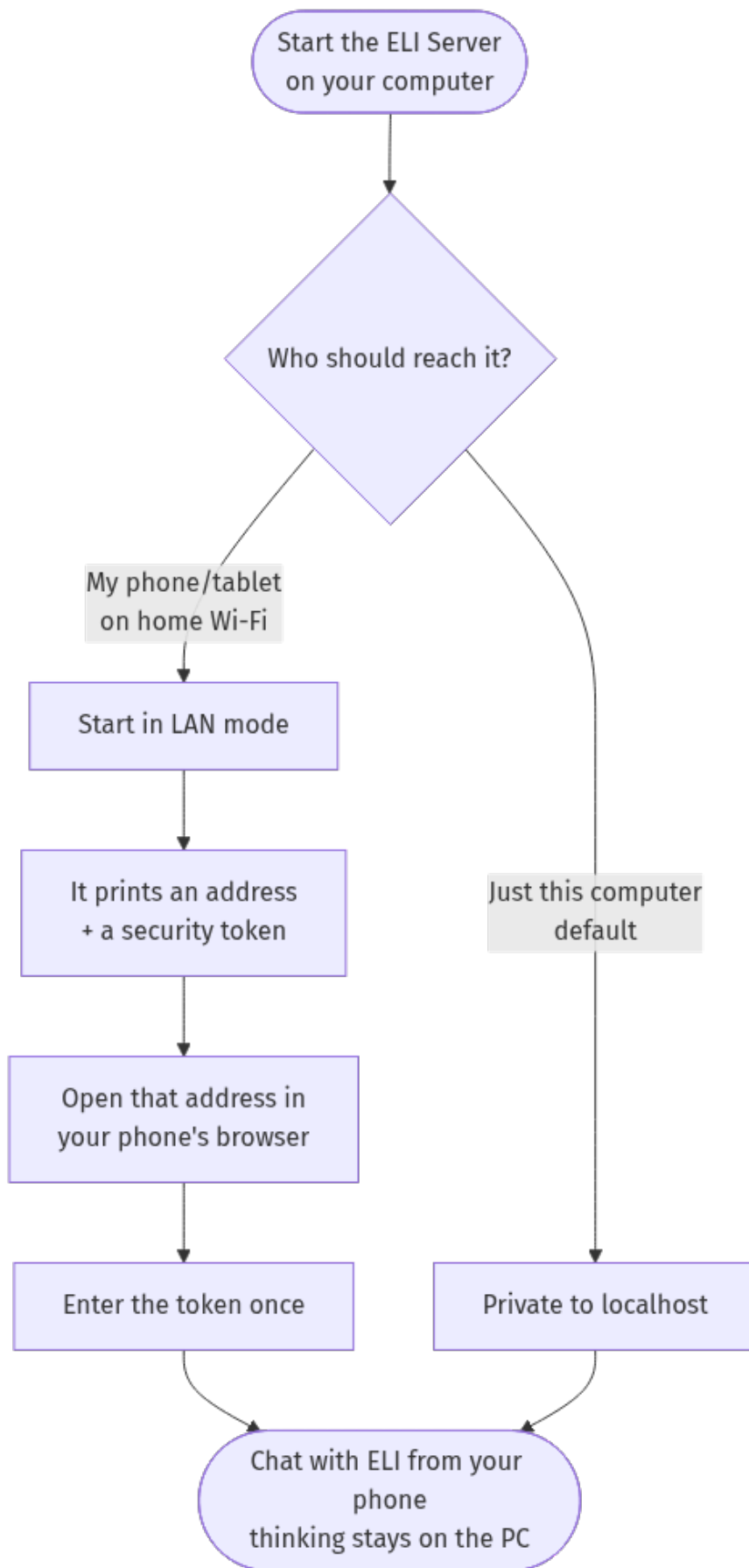
The first time you use vision, ELI may fetch a small vision model (you can pre-download it — §14). With a webcam set up, ELI even supports **eye-tracking control**: “enable gaze control”, “calibrate gaze”, then click where your eyes rest.

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### 13. Using ELI from your phone

[A bit technical] *Slightly technical, but only at setup.*

ELI includes a built-in **web app**. Start the server on your computer and chat with ELI from your **phone or tablet's browser** over your home Wi-Fi — while all the actual thinking stays on your computer.



- Start it from the desktop app — **Settings → Web Server** (a “this computer” button and a “phone / Wi-Fi” button that shows the exact link to open). Or run `scripts/eli_serve.sh` (Mac/Linux) / `scripts/eli_serve.ps1` (Windows). Full details: `docs/SERVER_AND_WEB_APP.md`.
- The web app is **installable** — your phone’s “Add to Home Screen” makes it open like an app, and the shell works offline. Light/dark themes; you can recolour the chat.

## What’s in the web app

It has a sidebar with several sections: - **Overview** — a live dashboard (clock, GPU/CPU/RAM gauges, status, recent activity). - **Chat** — streamed replies with markdown/code, saved sessions, stop/regenerate, and voice in/out. - **Commands** — a searchable list of everything ELI can do. - **Home** — ELI’s own **home-AI** (see below). - **System** — live, measured GPU/CPU/RAM/model telemetry. - **Research** — shared document corpora you can build and query **together**, with citations. - **Audit** — a tamper-evident log of every action. - **Admin** — (admin only) per-user activity, the risk-gate policy, and user management.

## Roles (optional)

By default the person on the computer is the operator (full admin). If others share it, you can create accounts with roles — **viewer** (read-only dashboards), **member** (use everything), **admin** (also the Admin console) — in the Admin tab. Each gets a one-time link; the audit trail then attributes actions to the right person.

## The Home tab — ELI as your home assistant

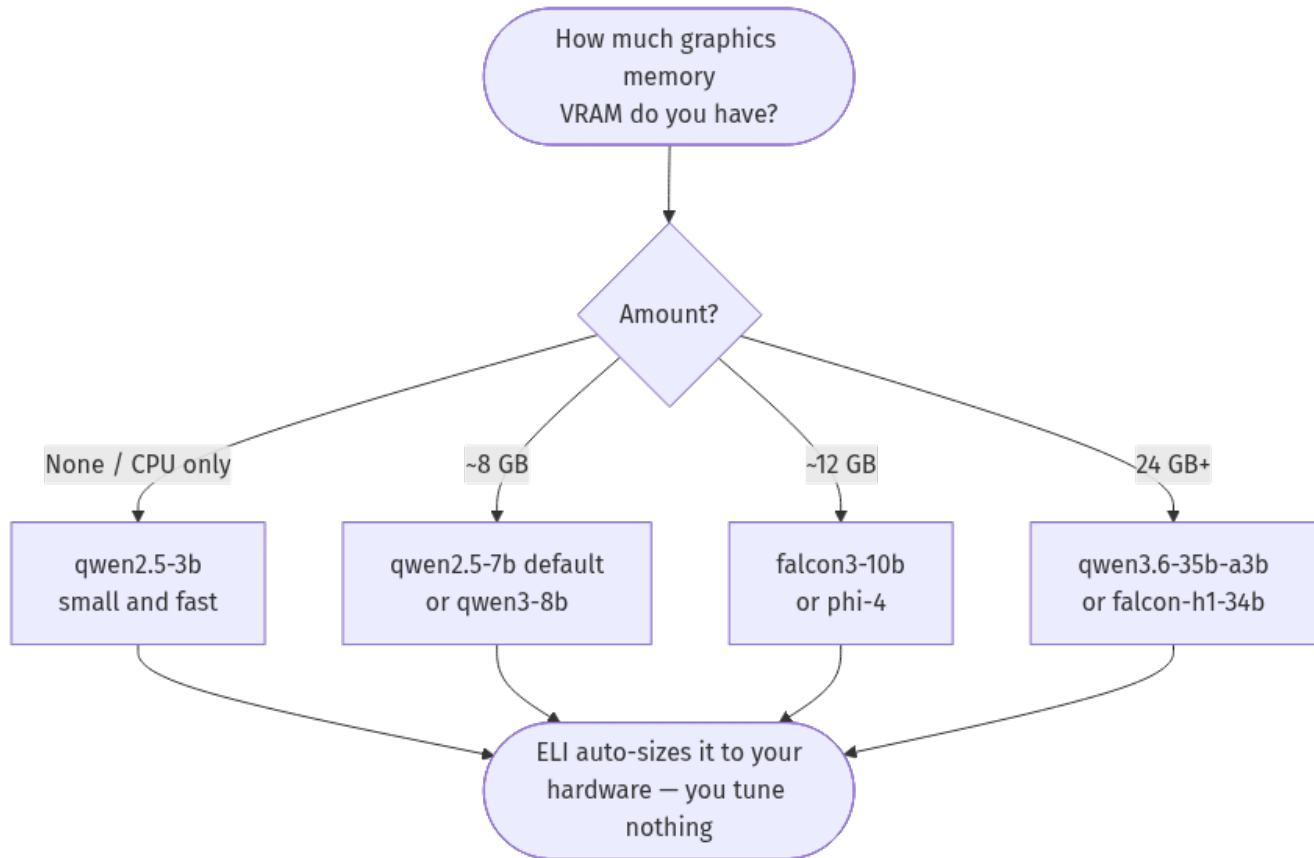
ELI controls smart devices **directly over MQTT** (ESPHome / Tasmota / Zigbee2MQTT) — no Home Assistant, no cloud: - **Find devices** on your network with one click (mDNS), or add one by its topics. - **Control** lights/switches, group them into **rooms**, and run room-wide on/off — from the app or by voice (“turn on the desk lamp”, “turn off the kitchen”). - **Scenes** — save a set of device states as “Movie mode” and activate it (by voice too). - **Automations** — *when* something happens (a time, **sunrise/sunset**, or a device turning on/off), *do* something (control a device or run a scene), optionally *only if* a condition holds. - ELI **learns** how you use your home and **suggests** automations you can accept with one tap.

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## 14. Models: picking, switching, and training your own

[A bit technical] The **model** is the “brain” ELI thinks with — a file on your computer. Bigger = smarter but needs more graphics memory (VRAM). ELI picks a suitable one at install; change it anytime.

## Which model should I get?



## Downloading

- **Pick from a menu (choose any number of models):** `python -m eli.core.model_download --choose` — this is what the installer uses; you tick the ones you want (or all / auto / none).
- Let ELI choose one best-fit for your hardware: `python -m eli.core.model_download --auto`
- See the menu without downloading: `python -m eli.core.model_download --list`
- Grab a specific one: `python -m eli.core.model_download qwen2.5-7b`

You're not limited to one — download several and switch between them anytime (§“Switching models”).

| Command name         | Model                              | Roughly needs          |
|----------------------|------------------------------------|------------------------|
| qwen2.5-3b           | small & fast                       | 4 GB graphics / or CPU |
| qwen2.5-7b (default) | great all-rounder                  | 8 GB graphics          |
| qwen3-8b             | newer, big context, good reasoning | 8 GB graphics          |
| falcon3-10b          | a step up                          | 12 GB graphics         |
| phi-4                | strong reasoning for its size      | 12 GB graphics         |
| qwen3.6-35b-a3b      | very capable (mixture-of-experts)  | 24 GB / or slow on CPU |
| falcon-h1-34b        | largest option                     | 24 GB / or slow on CPU |

You can also just **drop any .gguf file** into the `models/` folder and ELI finds it.

## Switching models

In **Settings** → **Model** (or “Model menu / Auto Detect”), pick the file. ELI sizes it to your hardware automatically — you don’t tune anything.

## [Advanced] Training your own ELI-flavoured model (advanced, optional)

You can fine-tune a model so it speaks in **ELI’s voice** out of the box.

-> **See blueprints/finetuning\_guide.md** — a full step-by-step from choosing a base model, through extracting a voice dataset and training, to producing a ready-to-load .gguf. Crucially, ELI’s **personality stays live** (never frozen into the model); training only shapes the *manner* of speech.

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## 15. How hard ELI thinks (reasoning modes)

[Anyone] ELI has five effort levels. Faster modes answer quickly; deeper modes think longer and use more of ELI’s internal helpers for hard problems.

| Mode            | Feel               | Use it for                              |
|-----------------|--------------------|-----------------------------------------|
| <b>quick</b>    | snappy             | greetings, simple commands, quick facts |
| <b>normal</b>   | balanced (default) | everyday questions and tasks            |
| <b>advanced</b> | more thorough      | tricky, multi-step asks                 |
| <b>research</b> | digs deep          | gathering and synthesising a lot        |
| <b>expert</b>   | maximum effort     | the hardest problems                    |

Ask “what reasoning mode are you in?” or “explain all your reasoning modes”. For most people the default is right — ELI also deepens on its own when a problem clearly needs it.

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## 16. Settings you’ll actually touch

[A bit technical] Most people never need these. The few that matter:

- **Model** — which brain ELI uses (\$14).
- **Voice** — wake word, text-to-speech on/off, microphone.
- **Hardware / startup** — ELI auto-tunes; advanced users can nudge how much graphics memory to hold back, or how much context to use. Leave alone unless you know why.
- **Privacy** — ELI is offline by default; this is where anything network-related lives.

Two graphics cards? ELI can use both — set in the GPU profile settings; single-card just works.

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## 17. Privacy & staying offline

[Anyone]

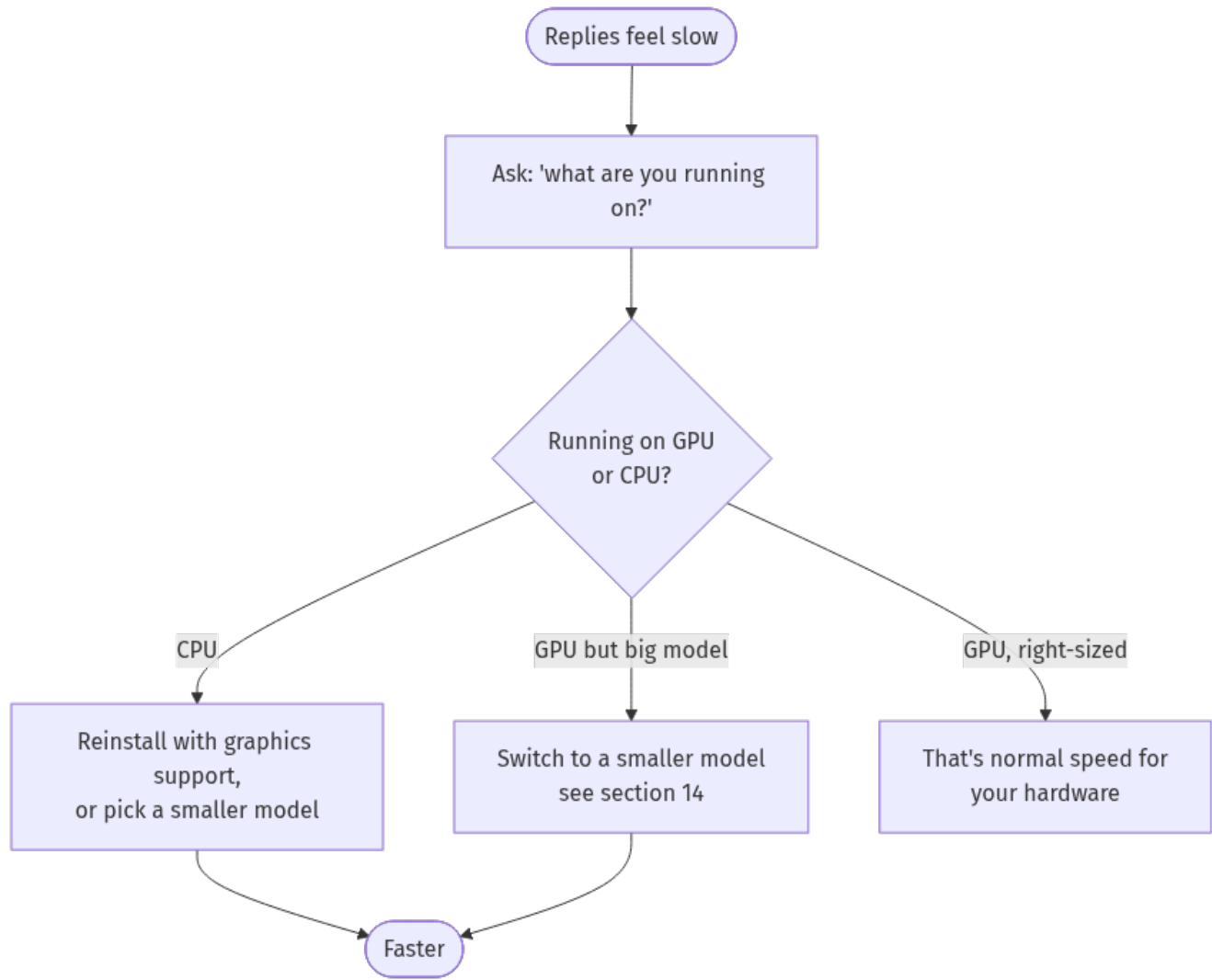
- **Nothing leaves your computer** unless you ask for something online (web search, news, weather, downloading a model). ELI tells you when it goes online.
  - **Your data is yours** — conversations, memories, files, profile live in local databases. Delete anytime.
  - **No accounts, no telemetry, no subscription.**
  - A fresh install is a **blank slate** — ELI starts knowing nothing about you.
-



# 18. Troubleshooting

[A bit technical] Most issues map to one of these. When in doubt, **ask ELI** — “run a full runtime audit” or “status” — it reports honestly.

## “ELI is replying very slowly”



## Everything else

| Problem                               | Try this                                                                                                                                |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Replies look like gibberish</b>    | A model whose context is too small. Pick a recommended one (§14); ELI sizes context automatically.                                      |
| <b>“I couldn’t find a model”</b>      | <code>python -m eli.core.model_download --auto</code> Normal on a fresh install. Ensure the embedder downloaded: <code>python -m</code> |
| <b>Memory/recall seems empty</b>      | <code>eli.core.model_download --aux</code>                                                                                              |
| <b>Voice/wake word not responding</b> | “run voice diagnostics”; check the microphone in Settings → Voice.                                                                      |

| Problem                                | Try this                                                                                 |
|----------------------------------------|------------------------------------------------------------------------------------------|
| <b>Won't go online for news/search</b> | By design (offline-first). The network is allowed per-request; check Settings → Privacy. |
| <b>A habit didn't run</b>              | Open <b>Habits</b> — confirm it's enabled and the time is right; or "show my habits".    |
| <b>Something seems broken</b>          | "run a full runtime audit" or "status".                                                  |

## 19. Quick phrasebook (cheat sheet)

**Everyday** - "open spotify and play lo-fi" · "pause" · "skip" - "what's on my screen?" · "summarise report.pdf" - "set an alarm for 7am" · "set a timer for 10 minutes" - "what's the news?" · "weather in Dublin" · "what time is it?"

**Memory & you** - "remember that ..." · "what do you know about me?" · "my name is Alex"

**Habits & proactive** - "show my habits" · "yes" / "no" (to an offer) · "morning report" · "start proactive mode"

**Writing & code** - "write a report on X" · "write a script to ..." · "fix the bugs in foo.py"

**Voice** - wake word "computer" + request · "change the wake word to athena" · "start dictation" · "say good morning" · "train my voice"

**Check on ELI** - "what are you running on?" · "status" · "what have you been working on?"

**Models (in a terminal)** - `python -m eli.core.model_download --list` · `python -m eli.core.model_download --auto`

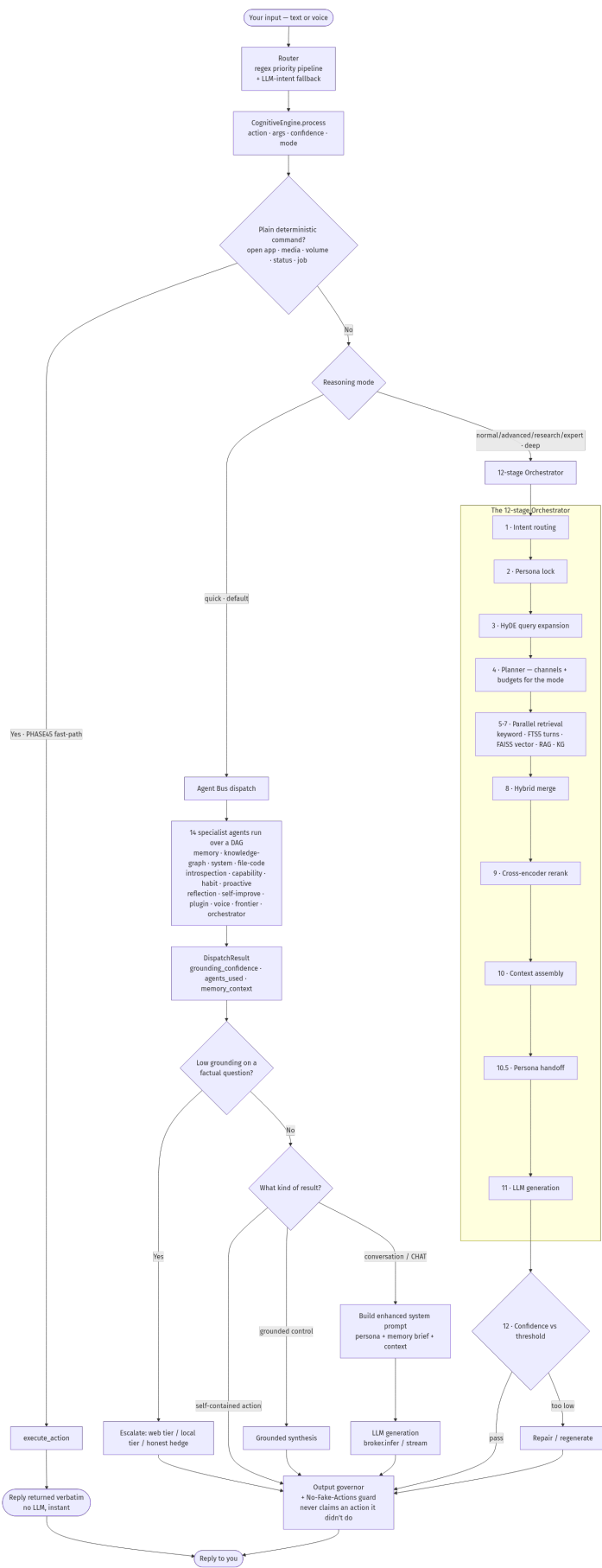
## 20. Glossary

| Term                      | Plain meaning                                                                       |
|---------------------------|-------------------------------------------------------------------------------------|
| <b>Model / GGUF</b>       | The "brain" file ELI thinks with. .gguf is just its file format.                    |
| <b>VRAM</b>               | Memory on your graphics card. More VRAM → you can run a bigger, smarter model.      |
| <b>Embedder</b>           | A tiny helper model that powers ELI's memory/recall. Installed automatically.       |
| <b>Context</b>            | How much of the conversation ELI can "hold in mind" at once. ELI sets this for you. |
| <b>Vision model</b>       | The model that lets ELI understand screens and images.                              |
| <b>Wake word</b>          | The phrase that makes ELI start listening (default "computer").                     |
| <b>Proactive mode</b>     | ELI doing helpful things on its own (briefings, suggestions).                       |
| <b>Habit</b>              | A routine ELI learned and you approved, run on a schedule.                          |
| <b>Fine-tuning</b>        | Optional advanced step to teach a model ELI's voice (\$14).                         |
| <b>Offline-by-default</b> | ELI never uses the internet unless you ask for something that needs it.             |

## 21. Appendix A — Under the hood: how ELI thinks (the full pipeline)

[Advanced] *For the curious / technical. You never need this to use ELI — but here is the complete, verified path from your words to ELI's reply.*

Every request runs through one funnel — `CognitiveEngine.process()` — which takes one of three paths depending on what you asked and the reasoning mode. The diagram is the **real** pipeline (verified against the running code): a deterministic fast-path for plain commands, a **quick** path via the Agent Bus (the default), and a **deep** 12-stage Orchestrator for hard work.



**The retrieval stages (5-7) are where memory plugs in** — that’s Appendix B. The two guarantees baked into the end of every path: the **Output governor** cleans the reply, and the **No-Fake-Actions guard** ensures ELI never claims it did something it didn’t.

## A.1 — The router (before any path is chosen)

Your words first hit the **router**. It runs a **regex-first priority pipeline**: an ordered set of fast, exact checks (web realtime lookups, media controls, file/PDF, memory, OS control, ...), each carefully shaped so it only fires on a real command — e.g. media controls need a whole word in command shape, so “metabolise” never triggers a media “tab”. If nothing matches, an **LLM intent classifier** is the fallback. The router emits {action, args, confidence} and hands off to `CognitiveEngine.process()`. The router also remembers the *last used path* so follow-ups (“do it again”, “the second one”) resolve correctly.

## A.2 — The three paths, and when each is taken

- **Fast-path (PHASE45) — no AI involved.** Plain, deterministic actions (open an app, pause music, set volume, report status, check a job) are executed directly and the result is returned **verbatim**. There’s no model call, so these are instant and can’t be “hallucinated”. This is why “pause” never produces a chatty paragraph.
- **Quick path (the default) — the Agent Bus.** Conversational and lightly-grounded requests go to the **Agent Bus**, which runs up to **14 specialist agents** in parallel over a dependency graph, each contributing evidence (memory hits, knowledge-graph facts, system readings, code search, capability info, ...). Their combined result carries a **grounding confidence** — an honest “is this actually backed by anything?” score.
- **Deep path — the 12-stage Orchestrator.** When you ask for depth (advanced/research/expert modes, or a hard question), ELI runs the full pipeline below: more retrieval, reranking, and a confidence check with automatic repair.

## A.3 — The Agent Bus, in full

The bus **selects a relevant subset** of agents for your intent (or fans out broadly when unsure), runs them concurrently, and **aggregates** their findings into a `DispatchResult`. Key agents and what they bring:

| Agent                                                                                         | Contributes                                                                                              |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Memory</b>                                                                                 | recalls facts/turns from SQLite + full-text + vector search                                              |
| <b>Knowledge-Graph System</b>                                                                 | entities and how they relate (who/what/where)<br>runs executor actions (web search, audits, status)      |
| <b>File-Code</b>                                                                              | searches the actual codebase for grounded answers about ELI itself                                       |
| <b>Introspection Capability</b>                                                               | live runtime/self state<br>what ELI can do (the capability manifest)                                     |
| <b>Habit / Proactive / Reflection / Self-Improve Plugin / Voice / Frontier / Orchestrator</b> | routines, signals, insights, fixes<br>plugin dispatch, speech state, deeper reasoning<br>hooks, planning |

After the bus returns, a **grounding-escalation hook** checks: *is this a factual question that came back weakly supported?* If so, ELI escalates — a web tier (if you allow online), a local tier, or an **honest hedge** (“I’m not certain, but...”) rather than a confident guess. Then the result is finalised: a self-contained action returns directly, a grounded control action gets a short grounded synthesis, and a conversation goes to generation with the full persona + memory context.

#### A.4 — The 12 stages, one by one (the deep path)

1. **Intent routing** — settle exactly what you're asking for.
2. **Persona lock** — fix ELI's identity/voice for this turn so it stays consistent.
3. **HyDE query expansion** — ELI drafts a *hypothetical answer* and searches with **that** too, which dramatically improves what memory retrieves (you find better matches by searching with an answer-shaped query, not just the bare question). Skipped for very short queries.
4. **Planner** — decide *which* retrieval channels to use and how big the budgets are, tuned to the reasoning mode (fast / balanced / deep). 5–7. **Parallel retrieval** — several searches run at once: keyword, **full-text (FTS5)** over your conversation history, **FAISS vector** (semantic) search, optional RAG, and the **knowledge graph**. (This is the seam where Appendix B's memory system feeds in.)
5. **Hybrid merge** — combine the hits from every channel into one candidate set.
6. **Cross-encoder rerank** — a smarter model re-scores the merged candidates for true relevance to your question, and the best rise to the top. (Skipped for non-conversational asks.)
7. **Context assembly** — build the grounded context block that will inform the answer. 10.5. **Persona handoff** — wrap that context with ELI's live persona brief for generation.
8. **LLM generation** — the model writes the reply (streaming or one-shot). For the deepest modes this is a two-part private-reasoning-then-condense step.
9. **Confidence check** — score the answer against a threshold; if it's too low, ELI **re-pairs/regenerates** rather than shipping a weak answer.

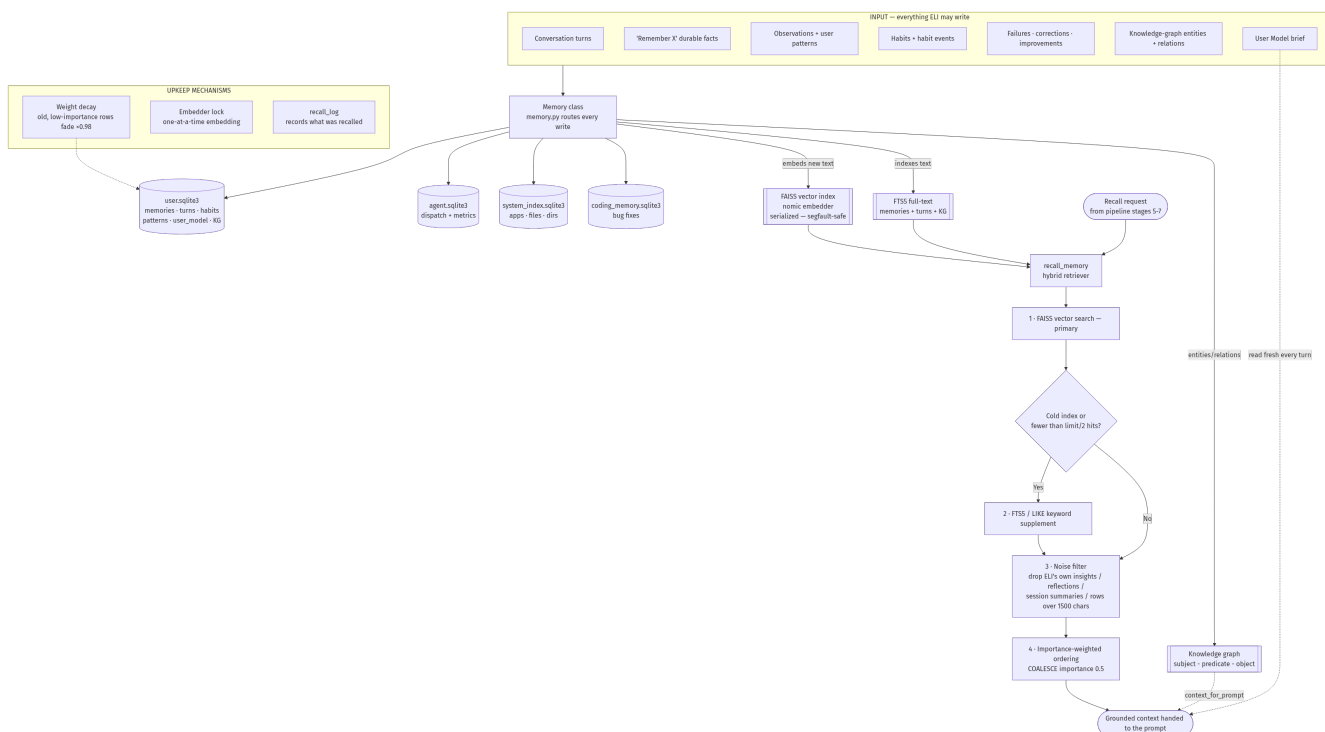
#### A.5 — The two end-of-path guarantees

Every path, no matter which, ends at: - **Output governor** — normalises and cleans the reply (strips artefacts, enforces formatting). - **No-Fake-Actions guard** — if ELI's reply *claims* it did something (played a song, fixed a file), the system verifies it actually happened; if not, the claim is replaced with the honest truth. This is enforced in code, not left to the model's goodwill.

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## 22. Appendix B — Under the hood: how ELI remembers (the full memory system)

[Advanced] *For the curious / technical. ELI's memory is a genuine hybrid — relational + full-text + vector + graph — all local. This is the complete input → storage → recall path (verified against the code).*



**Why it's trustworthy:** the **noise filter** stops ELI's own past replies and reflections from resurfacing as if they were *your* facts; the **embedder lock** keeps the vector engine from crashing under concurrent use; and **weight decay** is the gentle "forgetting" that lets stale topics fade while active ones stay fresh. Nothing here leaves your computer.

## B.1 — What ELI writes (the inputs)

Everything that flows into memory passes through one **Memory** module, which decides where each item belongs: - **Conversation turns** — what you said and what ELI replied, with timestamps. - **Durable facts** — anything you explicitly ask it to "remember". - **Observations & user patterns** — quiet signals about your interests, focus, and tone that build the living profile. - **Habits & habit events** — your repeated actions and the routines learned from them. - **Failures, corrections, improvements** — ELI's own mistakes and fixes, kept for self-learning (these are deliberately **excluded** from normal recall — see the noise filter). - **Knowledge-graph entities & relations** — named things and how they connect. - **User Model brief** — a compact, pre-rendered summary of who you are, refreshed over time.

## B.2 — Where it's stored (four local databases + three indexes)

Everything is plain local files on your machine — no server, no cloud: - **user.sqlite3** — the main store: memories, conversation turns, habits, patterns, the User Model, and the knowledge graph. - **agent.sqlite3** — ELI's internal agent activity and performance metrics. - **system\_index.sqlite3** — an index of your apps, executables, files, and folders (so "open the file manager" is instant). - **coding\_memory.sqlite3** — remembered code bug-fixes.

On top of the relational data sit three search indexes: - **FAISS vector index** — every stored text is turned into an "embedding" (a list of numbers capturing meaning) by a small local **nomic embedder**, so ELI can find things by *meaning*, not just exact words. Embedding is done **one-at-a-time** behind a lock (the embedder isn't thread-safe; serialising it prevents crashes). - **FTS5 full-text index** — fast exact/keyword search over memories, conversation turns, and the knowledge graph. - **Knowledge graph** — a subject → predicate → object web (e.g. *you* → *own* → *a Tesla*) that ELI can walk to answer relational questions.

### B.3 — How recall works (step by step)

When the cognition pipeline (stages 5–7) needs memory, it calls **recall\_memory**, a genuine hybrid retriever that runs in this order: 1. **Vector search first (FAISS)**. The semantic index is the primary path — it finds things that *mean* the same even if worded differently. 2. **Keyword supplement (FTS5/LIKE) — only if needed**. If the vector index is cold/empty or returns fewer than half the requested results (e.g. a very short query), a keyword search tops it up. (When another part of the system already did the vector search, this step is skipped to avoid double-searching.) 3. **Noise filter**. Results are scrubbed of ELI’s own output — assistant insights, reflections, session summaries, the “orchestrator” source, and any over-long blobs (>1500 chars). This is the safeguard that stops ELI’s past musings from masquerading as *your* facts. 4. **Importance-weighted ordering**. What survives is ranked by an importance score so the most significant memories surface first.

Alongside this, the **knowledge graph** contributes a lightweight relational context, and your **User Model brief** is read **fresh every single turn** — that’s why ELI stays in context about who you are without you repeating yourself.

### B.4 — Upkeep (the mechanisms that keep memory healthy)

- **Weight decay** — old, low-importance memories are gently faded (multiplied down over time), so abandoned topics naturally recede while active ones stay prominent. This is ELI’s “forgetting”.
- **Embedder serialization** — the single lock around the embedder that keeps the vector engine stable under concurrent access.
- **Recall log** — a record of what was retrieved, used for tuning and for honestly explaining *where* a piece of knowledge came from (“how do you know that?”).

### B.5 — Privacy, restated

Every database and index in this appendix is a local file under your user profile. There is no sync, no upload, no account. Delete the files (or use the **Memory** tab) and that knowledge is gone. A fresh install starts with the full structure but **zero** personal rows — a true blank slate.

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*That’s the whole assistant — inside and out. You don’t need any of the last two appendices to use ELI; just talk to it in plain English, and ask “what can you do?” whenever you’re curious. Everything here happens on your computer, for you, privately.*